

GridGuard - Milestone 1:

Project Scaffold

Before writing any code, carefully read:

- CLAUDE.md
- SPEC.md

These documents are the source of truth for this project.

Follow them exactly.

Do not make architectural changes or introduce technologies outside the documented stack.

If something is unclear, choose the simplest solution that remains consistent with both documents.

Objective

Create the initial repository scaffold for GridGuard.

This milestone establishes the project foundation only.

Do not implement application functionality.

The result should be a clean, modular project that builds successfully and is ready for future milestones.

Technology Stack

Use only the technologies defined in CLAUDE.md and SPEC.md.

Do not substitute frameworks or libraries.

Repository Structure

Create the folder structure exactly as defined in CLAUDE.md.

Do not add unnecessary folders or files.

Only create additional files when required by the selected frameworks.

Frontend

Initialize the frontend using:

- React (Vite)
- TypeScript
- Tailwind CSS
- shadcn/ui
- React Router

Create placeholder pages:

- Login
- Dashboard
- TransformerDetails

Create reusable component folders:

- common
- layout
- dashboard
- transformer
- alerts
- ui

Each page should compile successfully and contain only minimal placeholder content.

Backend

Initialize the backend using FastAPI.

Create placeholder modules for:

- api/routes
- auth
- database
- models
- schemas
- services
- utils

Inside the services folder create:

- generator.py
- predictor.py
- alert_manager.py
- ai_reporter.py

Do not implement business logic.

Project Setup

Configure:

Frontend

- Vite
- TypeScript
- Tailwind CSS
- shadcn/ui
- React Router

Backend

- FastAPI
- Python 3.12

Use:

- Node.js 22 LTS
- npm

Create:

- .gitignore
- README.md
- frontend/.env.example
- backend/.env.example

The README should briefly explain:

- Project purpose

- Technology stack
- Folder structure
- Local setup
- Development commands

Environment example files should contain placeholder variables only, matching the names defined in CLAUDE.md:

frontend/.env.example

- VITE_SUPABASE_URL
- VITE_SUPABASE_ANON_KEY

backend/.env.example

- SUPABASE_URL
 - SUPABASE_SERVICE_ROLE_KEY
 - CLAUDE_API_KEY
-

Placeholder Rules

Every placeholder should:

- compile successfully
- have meaningful names

- include a short TODO describing its future purpose

Avoid empty files whenever practical.

Do Not Implement

Do not implement:

- authentication logic
- Supabase integration
- database operations
- telemetry generation
- prediction logic
- alert logic
- dashboard functionality
- API endpoints
- responsive styling
- animations
- charts
- maps
- AI integrations
- machine learning
- mock telemetry

This milestone is for scaffolding only.

Before Writing Code

First provide:

1. A short implementation plan.
2. Any assumptions being made.
3. A brief explanation of the chosen project structure.

After presenting the plan, scaffold the repository.

Before Finishing

Verify that:

- frontend builds successfully
- backend starts successfully
- imports resolve correctly
- there are no broken references
- placeholder files contain TODO notes
- the project structure matches CLAUDE.md

Once all checks pass, initialize a git repository (if one does not already exist) and create a single initial commit with a clear message such as "Milestone 1: project scaffold".

Finally, summarize:

- folder structure
- frontend architecture
- backend architecture
- the initial commit
- readiness for Milestone 2 (Authentication)

Only finish once these checks have passed and the initial commit has been created.